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Department of Data Science
Course:- Statistical Programming Lab
Course Code:- 24CS04PR0404

Bank Customer Churn Prediction using Machine Learning

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IV Semester Section:-B4
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Bank Customer Churn Prediction using Machine Learning

Project Objective

The objective of this project is to build a machine learning model to predict whether a bank customer will churn. Multiple classification models are trained and compared to identify the best-performing model.

Introduction

Customer churn refers to customers leaving a service. In banking, predicting churn helps improve customer retention. Machine learning classification techniques can analyze customer data and predict whether a customer will leave the bank.

Project Methodology

1. Dataset Loading and Exploration

The dataset was loaded using pandas. Basic exploration was done using:

- head()
- info()
- describe()

2. Data PreprocessingRemoved unnecessary columns:

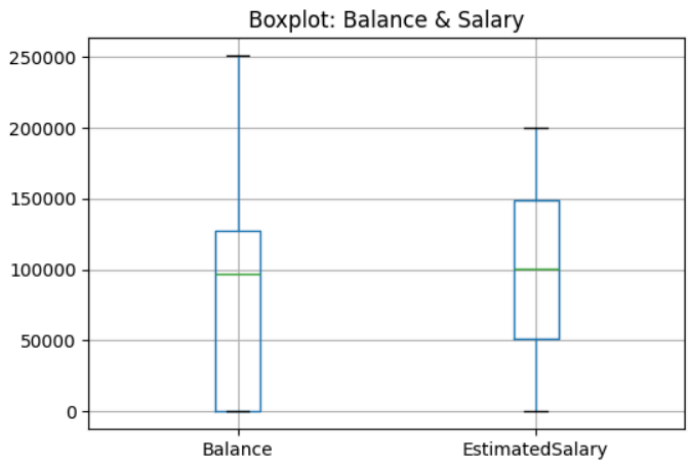
- Removed unnecessary columns:
    - RowNumber
    - CustomerId
    - Surname
- Converted categorical data (Geography, Gender) into numerical format using encoding.

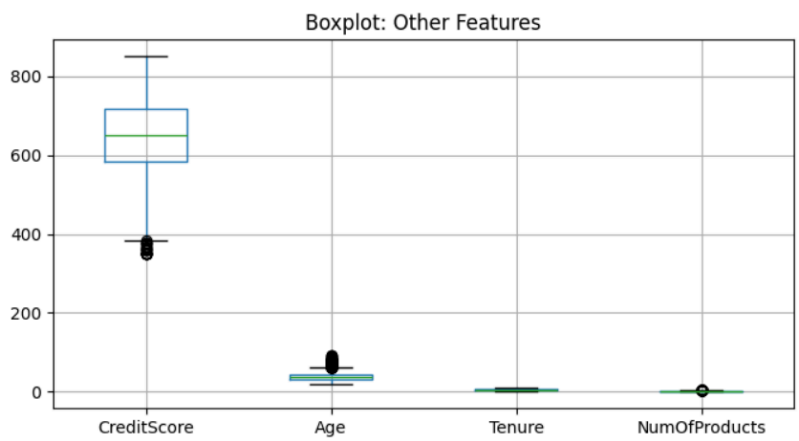
3. Feature Extraction

- Independent variables: Customer features
- Dependent variable: Exited (churn)

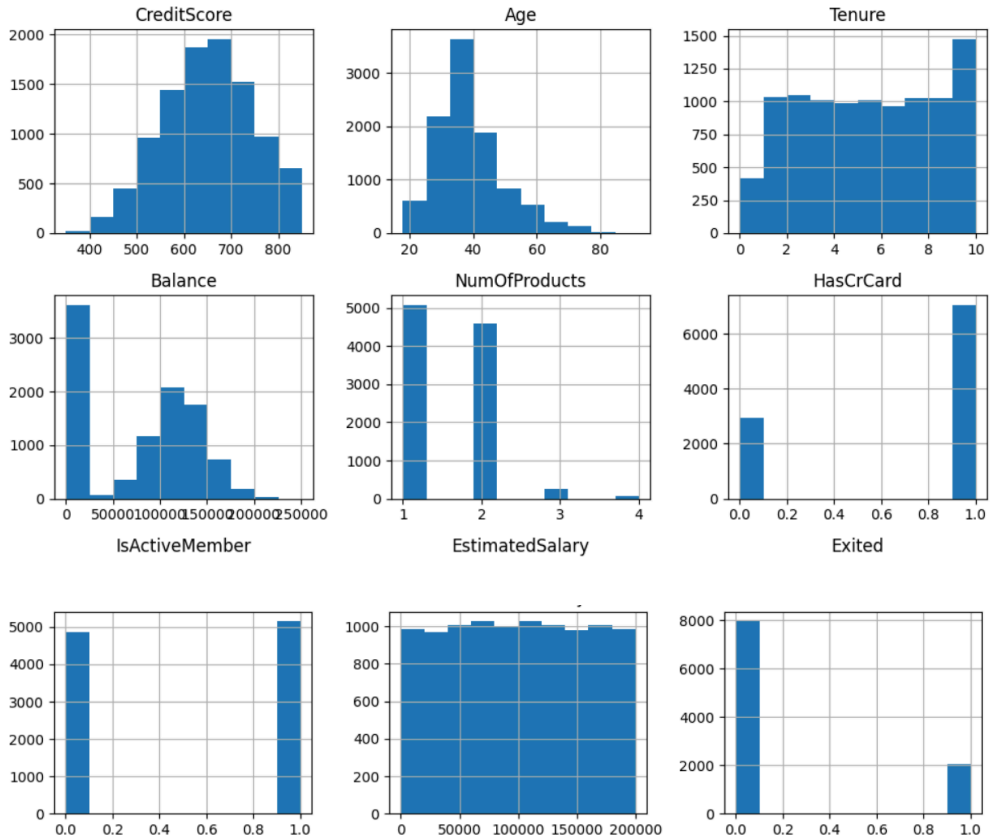
4. Data Visualization

- Box Plot:- Due to varying scales of features, separate boxplots were used to better visualize the distribution and detect outliers.

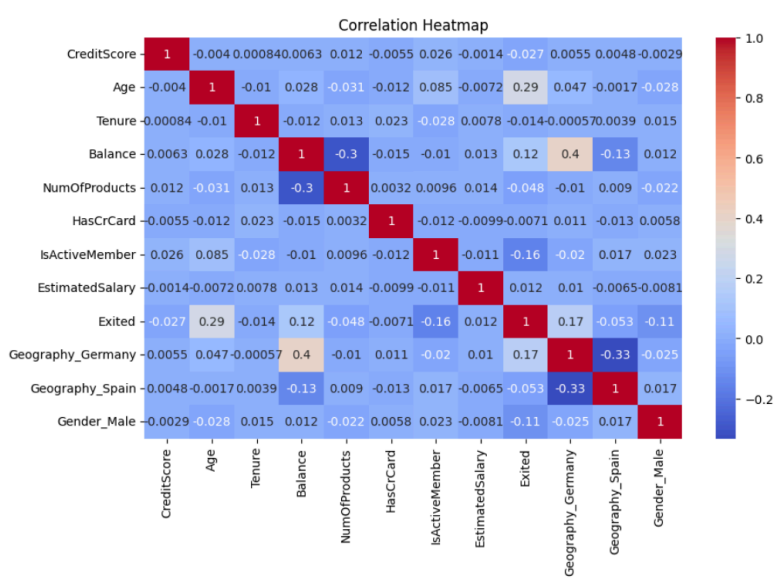




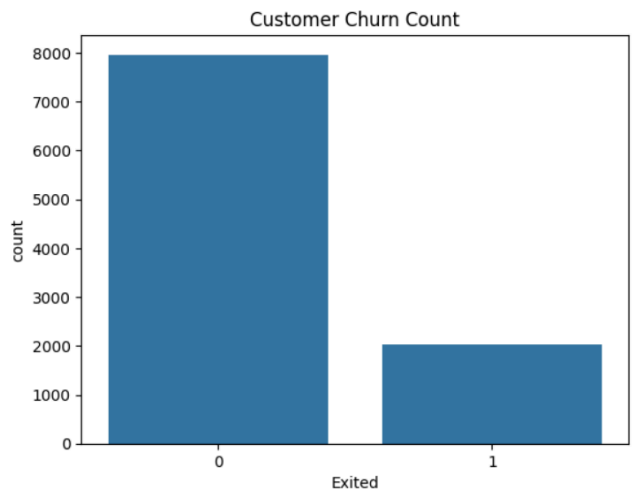
- Histogram:-Histograms were used to visualize the distribution of numerical features.



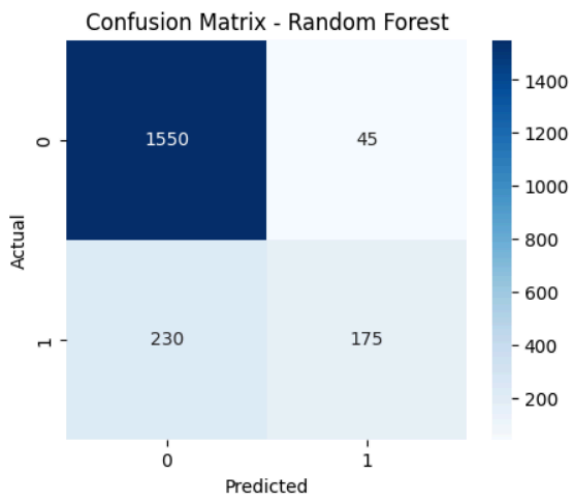
- Correlation Heatmap:- A correlation heatmap was used to identify relationships between variables.



- Count Plot:-Count plot shows distribution of churned and non-churned customers.



- Confusion Matrix:- The confusion matrix shows the performance of the model in correctly classifying churn and non-churn customers



5. Training Multiple Classification Models

The following models were trained:

- Logistic Regression

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest

6. Performance Evaluation and Model Comparison

Models were evaluated using:

- Accuracy
- Confusion Matrix

7. Best Model Selection

The model with the highest accuracy was selected as the best model.

8. GUI Implementation

A GUI was implemented using Streamlit to allow users to input data and get churn prediction results.

Project Implementation

The project was implemented in Python using Google Colab. Libraries used include:

- pandas
- numpy
- scikit-learn
- streamlit

Complete Code

```
import pandas as pd


df = pd.read_csv("Churn_Modelling.csv")

df.head()

df.info()

df.describe()
```

```
df = df.drop(['RowNumber','CustomerId','Surname'], axis=1)

df = pd.get_dummies(df, drop_first=True)

df.head()

X = df.drop('Exited', axis=1)

y = df['Exited']

from sklearn.model_selection import train_test_split


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)


import matplotlib.pyplot as plt


# Big scale features

df[['Balance', 'EstimatedSalary']].boxplot(figsize=(6,4))

plt.title("Boxplot: Balance & Salary")

plt.show()


# Small scale features

df[['CreditScore', 'Age', 'Tenure', 'NumOfProducts']].boxplot(figsize=(8,4))

plt.title("Boxplot: Other Features")

plt.show()


import matplotlib.pyplot as plt
```

```
df.hist(figsize=(10,8))
plt.tight_layout()
plt.show()

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

import seaborn as sns

sns.countplot(x='Exited', data=df)
plt.title("Customer Churn Count")
plt.show()

import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix

y_pred = rf.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Random Forest")

plt.show()


from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

lr = LogisticRegression(max_iter=1000)
knn = KNeighborsClassifier()
dt = DecisionTreeClassifier()
rf = RandomForestClassifier()
```

```
lr.fit(X_train, y_train)

knn.fit(X_train, y_train)

dt.fit(X_train, y_train)

rf.fit(X_train, y_train)


from sklearn.metrics import accuracy_score, confusion_matrix


models = {

    'Logistic Regression': lr,

    'KNN': knn,

    'Decision Tree': dt,

    'Random Forest': rf

}


for name, model in models.items():

    pred = model.predict(X_test)

    print(name)

    print("Accuracy:", accuracy_score(y_test, pred))

    print("Confusion Matrix:\n", confusion_matrix(y_test, pred))

    print("----------------------")


import pickle

pickle.dump(rf, open("model.pkl", "wb"))
```

```
!pip install gradio

import gradio as gr

import pickle

import numpy as np


# Load model

model = pickle.load(open("model.pkl", "rb"))


# Prediction function

def predict(credit, age, tenure, balance, products, card, active, salary, germany, spain, male):

    try:

        input_data = np.array([[credit, age, tenure, balance, products, card, active, salary, germany,
spain, male]])

        result = model.predict(input_data)


        if result[0] == 1:

            return "Customer will Churn"

        else:

            return "Customer will NOT Churn"

    except:

        return "Error in prediction"


# Interface
```

```
interface = gr.Interface(
    fn=predict,
    inputs=[
        gr.Number(label="Credit Score"),
        gr.Number(label="Age"),
        gr.Number(label="Tenure"),
        gr.Number(label="Balance"),
        gr.Number(label="Number of Products"),
        gr.Radio([0,1], label="Has Credit Card (0=No, 1=Yes)"),
        gr.Radio([0,1], label="Is Active Member (0=No, 1=Yes)"),
        gr.Number(label="Estimated Salary"),
        gr.Radio([0,1], label="Geography: Germany (1=Yes, 0=No)"),
        gr.Radio([0,1], label="Geography: Spain (1=Yes, 0=No)"),
        gr.Radio([0,1], label="Gender: Male (1=Yes, 0=No)")
    ],
    outputs="text",
    title="Bank Customer Churn Prediction"
)

interface.launch()
```

Results and Discussion

Model Accuracy Table

Model	Accuracy
Logistic Regression	81.75
KNN	75.65
Decision Tree	78.75
Random Forest	86.1

Best Model

Random Forest achieved the highest accuracy (86.1) and was selected as the best model.

Confusion Matrix Explanation

The confusion matrix shows that the model correctly predicts most non-churn customers, while some churn cases are misclassified due to imbalance in the dataset.

Output Screenshots

```python
import pandas as pd

df = pd.read_csv("Churn_Modelling.csv")
df.head()
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

Data preprocessing Output

```
df = df.drop(['RowNumber','CustomerId','Surname'], axis=1)
df = pd.get_dummies(df, drop_first=True)
df.head()
```

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited	Geography_Germany	Geography_Spain	Gender_Male
0	619	42	2	0.00	1	1	1	101348.88	1	False	False	False
1	608	41	1	83807.86	1	0	1	112542.58	0	False	True	False
2	502	42	8	159660.80	3	1	0	113931.57	1	False	False	False
3	699	39	1	0.00	2	0	0	93826.63	0	False	False	False
4	850	43	2	125510.82	1	1	1	79084.10	0	False	True	False

Model Accuracy Results and Confusion matrix

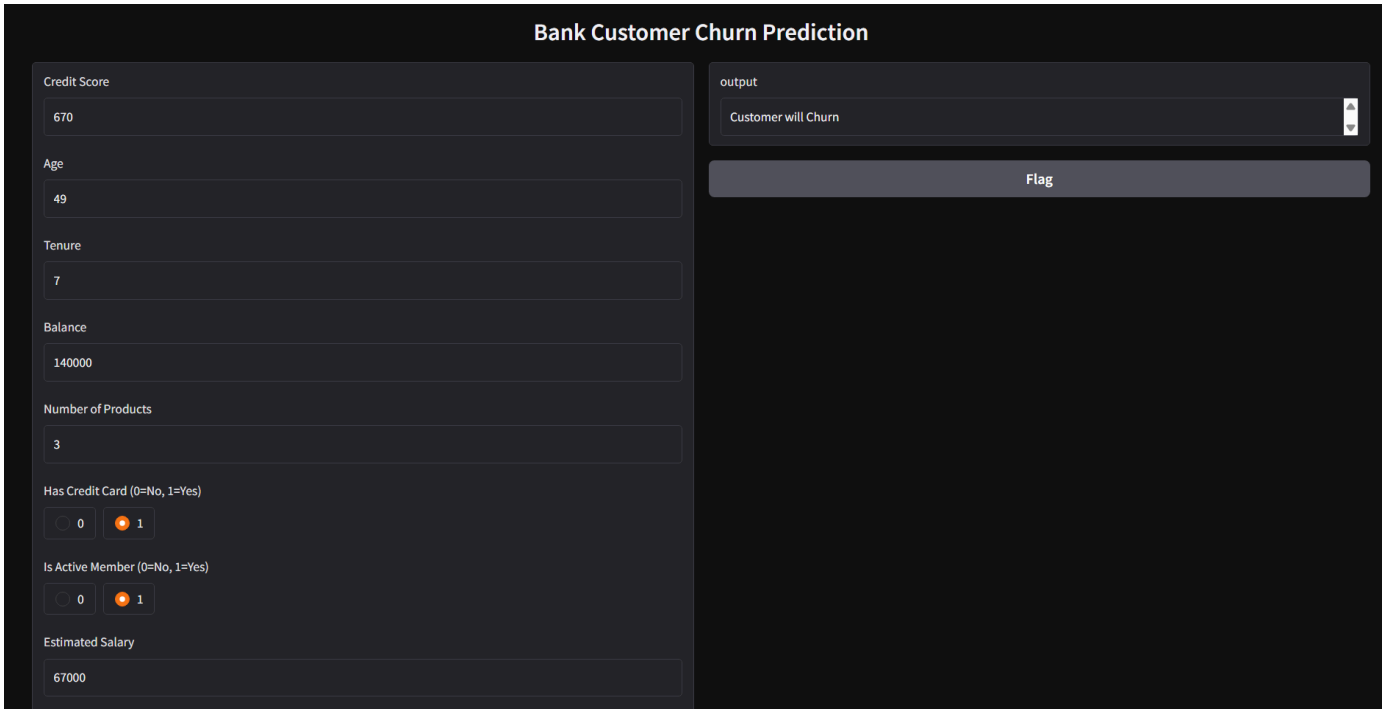
```
Logistic Regression
Accuracy: 0.8175
Confusion Matrix:
 [[1534   60]
 [ 305  101]]
----------------------
KNN
Accuracy: 0.7565
Confusion Matrix:
 [[1477  117]
 [ 370   36]]
----------------------
Decision Tree
Accuracy: 0.7875
Confusion Matrix:
 [[1366  228]
 [ 197  209]]
----------------------
Random Forest
Accuracy: 0.861
Confusion Matrix:
 [[1535   59]
 [ 219  187]]
----------------------
```

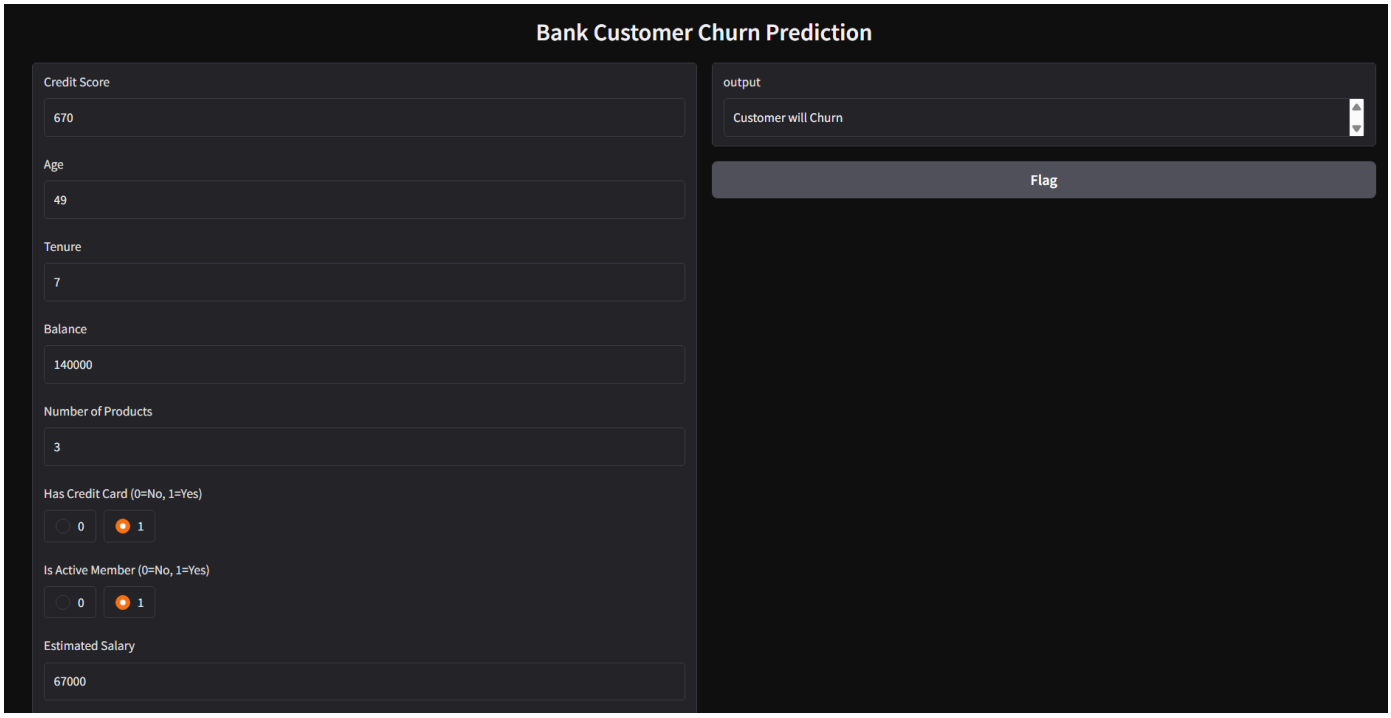
Best Model Output

```
Logistic Regression   KNN
Accuracy: 0.8175      Accuracy: 0.7565
---------------------- ----------------------

Decision Tree         Random Forest
Accuracy: 0.7875      Accuracy: 0.861
```

GUI Output:-


Bank Customer Churn Prediction
Credit Score
670
Age
49
Tenure
7
Balance
140000
Number of Products
3
Has Credit Card (0=No, 1=Yes)
0
1
Is Active Member (0=No, 1=Yes)
0
1
Estimated Salary
67000
output
Customer will Churn
Flag


Bank Customer Churn Prediction
Credit Score
670
Age
49
Tenure
7
Balance
140000
Number of Products
3
Has Credit Card (0=No, 1=Yes)
0
1
Is Active Member (0=No, 1=Yes)
0
1
Estimated Salary
67000
output
Customer will Churn
Flag

8. Conclusion:-

This project successfully developed a machine learning model to predict customer churn using the given dataset. The data was preprocessed by removing unnecessary features and converting categorical variables into numerical form. Multiple classification models, including Logistic Regression, KNN, Decision Tree, and Random Forest, were trained and evaluated.

Among these models, **Random Forest achieved the highest accuracy**, making it the most suitable model for this problem. The confusion matrix analysis showed that the model performs well in identifying non-churn customers, while some churn cases are misclassified due to class imbalance in the dataset.

A graphical user interface was implemented using Gradio to allow users to input customer details and obtain predictions. Overall, the project demonstrates how machine learning techniques can be applied to real-world problems like customer retention in banking.

9. Github Link:-Churn Prediction Project